

Anthropogenic Surface Disturbance Mapping: General Procedures

The BEACONs Project

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Overview

This document provides the basic information needed to undertake an anthropogenic surface disturbance mapping project within an area of interest in the Yukon (e.g., a watershed, caribou range, or other ecological and administrative boundaries) and, with minor modifications, in nearby jurisdictions (e.g., British Columbia). For additional information, we refer technicians to the accompanying documentation provided by the Yukon Government¹. For other jurisdictions, we have provided boxes highlighting additional considerations. The goal of a mapping project is to produce two GIS data layers for a given project area: one containing linear surface disturbances and the other containing areal (polygonal) surface disturbances. Anthropogenic surface disturbances include any feature created by human activities that is visible on satellite or aerial imagery.

This document contains the following sections:

- Project data package - A description of the datasets and digitizing template files that are provided in the data package, including a description of linear and polygonal attributes.
- Digitization methods - Detailed procedures for mapping linear and polygonal disturbances, and assigning attributes to each disturbance.
- Two appendices describing 1) industry and disturbance types and 2) a project checklist.

¹ Powell, K. 2022. Anthropogenic surface disturbance mapping in the Yukon. Standards and guidelines for contractors. Fish and Wildlife Branch, Environment Yukon.

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1. Project data package

This section explains how to set up a disturbance mapping project using the open source software QGIS. The setup process produces a QGIS project file and a project data package with the core layers needed for digitizing and validation.

A complete project typically includes the following components:

- a study area boundary layer;
- empty or existing linear and areal disturbance layers;
- two grids that intersect the study area: 2 x 2 km and 10 x 10 km;
- satellite imagery web services; and
- ancillary datasets to support interpretation and digitizing.

There are two pathways, detailed in Section 1.1, for creating a project file and data package. The appropriate pathway depends on whether your area of interest (AOI) falls inside or outside the boundary where compiled disturbance data are available, as shown in Figure 1. Both pathways include the following steps that are described in more detail below:

1. Acquire or create an area of interest (AOI) map for your study region.
2. Run the Geopackage Creator app and download disturbance and ancillary data for your AOI. If none are available, download the empty linear and areal disturbance layers from GitHub.
3. Create a QGIS project file with your AOI, disturbance layers, ancillary layers, and web services.

Supporting material on GitHub

The disturbance mapping GitHub repository contains manuals, QGIS template files, and validation tools needed to support a mapping project. You can view the repository on [GitHub](#). To download a local copy, select the green “Code” button and choose “Download ZIP”.

The repository is organized into the following folders and files:

- **root folder:** Includes a README file that describes the repository contents.
- **manuals:** Contains PDF versions of the manuals, including `general_procedures.pdf` and `temporal_mapping_simplified.pdf`.
- **template:** Includes sample files that you can use for your project.
 - Two QGIS project files with satellite web services added:
 - `digitizing_project1.qgz` can be used if your AOI contains disturbances data that can be downloaded using the [Geopackage Creator](#) app.
 - `digitizing_project2.qgz` can be used if your AOI has no available disturbance data, in which case it will load the `data_package.gpkg` file, which contains empty linear and areal disturbance layers.
 - A geopackage file with empty linear and areal disturbance layers (`data_package.gpkg`) in case there is no disturbance data in your study area.
 - A list of disturbance types and industries (`disturbance_types.csv`).
- **validation:** Contains a Shiny app to assist in viewing mapping results, performing attribute validation, and editing attributes. It is best run from this [link](#).



Figure 1. Boundary of datasets available within the Geopackage Creator app. If your AOI is within this area, use Option 1 below to create your project data package, otherwise use Option 2.

1.1 Creating a data package

To start a disturbance mapping project, create a geopackage file that contains the layers needed for digitizing and review. The workflow depends on whether your area of interest (AOI) falls within the dataset boundary shown in Figure 1.

Option 1: AOI is within the Figure 1 boundary

1. Prepare an AOI geopackage or shapefile. The AOI should contain one polygon in the coordinate reference system (CRS) you will use for digitizing, such as Yukon Albers Equal Area (EPSG:3578).
2. Open the AOI in the [GeoPackage Creator](#) and select the required layers.
 - a. linear_disturbance and areal_disturbance
 - b. grid_2x2km and grid_10x10km
 - c. any available ancillary layers that may help with digitizing

3. Save the AOI and selected layers in a geopackage named data_package.gpkg.
4. Download digitizing_project1.qgz from the template directory on GitHub.
5. Create a project directory on your computer for digitizing.
6. Place digitizing_project1.qgz and data_package.gpkg in the project directory.
7. Open digitizing_project1.qgz in QGIS and confirm that the disturbance datasets, ancillary datasets, and satellite imagery are available.
8. Begin digitizing.

Option 2: AOI is outside the Figure 1 boundary

1. Prepare an AOI geopackage or shapefile. The AOI should contain one polygon in the coordinate reference system (CRS) you will use for digitizing, such as Yukon Albers Equal Area (EPSG:3578).
2. Download digitizing_project2.qgz and data_package.gpkg from the template directory on GitHub.
3. Create a project directory on your computer for digitizing.
4. Place digitizing_project2.qgz and data_package.gpkg in the project directory.
5. Open data_package.gpkg in the Disturbance Validation app.
6. Use the app to create the 2 x 2 km and 10 x 10 km grids.
7. Save the new grids and all required layers in data_package.gpkg.
8. Open digitizing_project2.qgz in QGIS and confirm that the disturbance datasets, ancillary datasets, and satellite imagery are available.
9. Begin digitizing.

1.2 Map layers and web services

After creating the project directory, QGIS project file, and data_package.gpkg described in Section 1.1, the next step is to confirm that the project contains the layers and web services needed for interpretation, digitizing, and validation. The data package provides the core working layers, while web services supply the imagery and thematic reference information used to identify and classify disturbance features. This section describes the main map layers included in a typical project, identifies additional layers that may be useful, and provides guidance on using web services in QGIS or ArcGIS.

1.2.1 Map layers

Table 1 summarizes the layers commonly included in a disturbance mapping project. Some layers are created or downloaded during the data-package workflow, while others may be added as supporting information depending on the AOI, jurisdiction, and interpretation needs. Together, these layers help technicians define the study area, track mapping progress, avoid duplicating previously mapped features, and distinguish anthropogenic disturbances from natural features or unrelated land-cover patterns.

Table 1. Map layers commonly used in the project data package, including digitizing templates, reference layers, existing disturbance layers, and ancillary layers. Additional layers of interest can be found [here](#).

Group	Layer name	App ¹	Description
Digitizing templates	bp_poly	no	The layer to digitize in, for digitizing areal features like mines, road pull outs, campsites etc. These are empty but should contain all req'd attributes
	bp_line	no	The layer to digitize in, for digitizing linear features like roads, trails, seismic lines etc.
Reference layers	studyarea	user	Boundary of project area of interest (AOI) provided by user
	grid_10x10k	?	Large grid for managing disturbance mapping project
	grid_2x2k	?	Fine grid for tracking digitizing progress
	linear_disturbance	yes	YG anthropogenic linear disturbances; includes the national road network (NRN) and national railway network (NRWN) from Canvec
	areal_disturbance	yes	YG anthropogenic areal disturbances
Disturbance layers	yg_mapping_extent	no	YG spatial extent of anthropogenic surface disturbance mapping
	canada_access_2010	yes	Canada-wide surface disturbance map from 2010. May useful in areas with no previous disturbance mapping
	fire_history	?	Polygons of past fires. Sometimes cut blocks and fires can look similar
	burned_areas	?	National burned areas composite
	quartz_claims_50k	yes	Surface disturbances inside these claims areas are likely associated with quartz mining
Ancillary data	placer_claims_50k	yes	Surface disturbances inside these claims areas are likely associated with placer mining
	watercourses_50k	yes	1:50,000 watercourses from Canvec (rivers can sometimes look like trails)
	historic_silviculture	?	These are the specific silviculture activities that took place within an opening in order to meet silviculture obligations after a disturbance

¹ Indicates map layers that can be clipped to an AOI using the Geopackage Creator app.

Box 1. Other Jurisdictions

We have compiled a list of websites, disturbance datasets, supporting layers and satellite imagery for different regions from the Global scale, to Canada-wide, to datasets specific to the Yukon Territory, British Columbia, Alberta and the Northwest Territories. This resource is available [here](#) and through our Github

repository. Users should be aware that these datasets are not managed by BEACONS Project and may occasionally be retired, updated or moved.

1.2.1 Web services

In addition to the map layers stored in the project data package, most mapping projects require web services that can be viewed directly in QGIS or ArcGIS. These services provide the high-resolution imagery used to identify disturbance features, as well as thematic reference layers that help interpret their likely source or classification. For example, imagery services can be used as the primary digitizing backdrop, while forestry, hydrology, mining-claim, or other contextual layers can help distinguish anthropogenic disturbances from natural features and support consistent attribute assignment. The BEACONS Project maintains a growing list of global, national, provincial, and territorial web services that may be useful for disturbance mapping. These include imagery services such as ESRI and Google imagery, national reference layers such as hydrology, and jurisdiction-specific services such as satellite imagery or forestry datasets. The current list is provided in this [Google sheet](#), along with other publicly available datasets that may support project setup, interpretation, or validation.

For Yukon projects, the recommended primary imagery source is the SPOT composite at 150 cm resolution, which provides a consistent basis for digitizing both linear and areal disturbances. This service is included in the template QGIS project files. If a web service is missing from a project file, or if additional imagery or reference layers are needed for a specific AOI, add the service manually and confirm that it displays correctly before digitizing.

General instructions for adding web services are available from the following QGIS and ArcGIS documentation:

- [QGIS - Web Mapping Services](#)
- [ArcGIS - Adding map services](#)

1.3 Disturbance attributes

Following the setup of map layers and web services in Section 1.2, the next step is to understand the attribute structure associated with the disturbance layers. These attributes define how each mapped feature is described, classified, and validated, and they ensure consistency across projects and users.

The attribute schema used in the digitizing layers (either the empty bp_line and bp_poly layers, or the existing linear_disturbance and areal_disturbance layers) is based on the Yukon Government standard for anthropogenic surface disturbance mapping. This standardized structure allows newly digitized features to be integrated with existing datasets and supports downstream analysis and validation.

Table 2 summarizes the attributes used for both linear and areal disturbances, including their data types, domains, and descriptions. While many attributes apply to both geometry types, some (shown in bold in the table) are specific to linear features.

Two of the most important attributes for classification are:

- **TYPE_INDUSTRY** – identifies the sector responsible for the disturbance (e.g., mining, forestry, transportation); and
- **TYPE_DISTURBANCE** – provides a more detailed description of the specific feature type (e.g., seismic line, cutblock, airstrip).

The full set of allowable values for these attributes is provided in Appendix 1. Together, these fields form the basis for consistent classification of disturbance features across the project.

Several additional attributes capture information about how the feature was digitized and interpreted. For example:

- **DATA_SOURCE**, **IMAGE_NAME**, **IMAGE_DATE**, and related fields document the imagery used;
- **SCALE_CAPTURED** records the digitizing scale; and
- **FEATURE_VISIBILITY** indicates whether a feature was directly observed or interpolated from imagery.

Uncertainty or interpretation decisions can be recorded using the **NOTES** field. Features that require further review—either during validation or through field verification—should be flagged using the **FLAG** attribute (value = 1), with supporting details included in **NOTES**.

Additional guidance on how and when to populate specific attributes is provided in Section 2 (Digitization methods), where attribute assignment is described in the context of mapping workflows.

Table 2. Attribute schema for linear and areal disturbance layers in the BEACONS digitizing templates, including field names, data types, domains, and descriptions. Attributes in bold apply only to linear features, and BEACONS-specific fields are identified in blue.

Attribute	Data Type	Domains	Description
REF_ID	Text (100)		Unique feature reference ID - leave blank
DATABASE	Text (12)	Historic, Most Recent, Retired	Sub-database to which the feature belongs
TYPE_INDUSTRY	Text (20)	See Table 3	Major classification of disturbance feature by industry
TYPE_DISTURBANCE	Text (30)	See Table 3	Sub classification of disturbance feature (see Table 3)
WIDTH_M	Double		Width of feature in metres to the nearest whole number
WIDTH_CLASS	Text (5)	HIGH, MED, LOW	Width of feature by classification (no longer used)
SCALE_CAPTURED	Long		Scale at which the feature was digitized, generally 1:5000 (value=5000)

DATA_SOURCE	Text (10)	Imagery, GPS, NRN, NRW	Digitized from imagery, captured by GPS, adapted from the NRN or NRW, or obtained by other means
IMAGE_NAME	Text (100)		Filename of source imagery (e.g., SPOT 2020-2021 Mosaic / ESRI World Imagery)
IMAGE_DATE	Date		Date that imagery was captured
IMAGE_RESOLUTION	Double		Resolution of source imagery in metres
IMAGE_SENSOR	Text (35)		Name of sensor that captured source imagery
CREATED_BY	Text (25)	USER	User that created/digitized a feature
CREATED_DATE	Date		Date feature was created/digitized
FEATURE_VISIBILITY	Text (25)	VISIBLE, INTERPOLATED	Indicates whether the feature is visible and interpreted from the imagery, or if it was interpolated because the feature was obscured, but known to be present.
DISTURBANCE_YEAR	Integer (4)		Year that the disturbance appeared on the landscape
FLAG	Short Integer (1)		Feature is a high priority for validation (mark with value 1)
FLAG_YG	Short integer (8)	REF_ID from linear_disturbances or areal_disturbances	YG Feature is flagged for consideration for removal from the YG database
VHR_ASSIST	Text (10)	ESRI, Google, Bing	Name of VHR imagery, if any, used to aid in digitizing
NOTES	Text (200)		Notes

Box 2. Other Jurisdictions

Existing surface disturbance datasets for jurisdictions outside the Yukon Territory will use a different classification system for disturbance features. Users working in other jurisdictions must ensure to review the attribute table of existing datasets, and decide whether to simplify the BP_line and BP_poly attribute tables to match, or review and update existing datasets. This is an important consideration to ensure the existing and new datasets can be merged later on in the workflow. This decision can be made based on available project resources and the requirements of subsequent analyses that will rely on inputs from the final surface disturbance database.

2. Digitization methods

2.1 General approach

The following bullets provide important information about the general approach used to digitize linear and polygonal surface disturbances:

- **Previously mapped surface disturbances**
 - Each project file contains existing linear and polygonal disturbance layers (linear_disturbances and areal_disturbances which, at a minimum, includes the national road and railway networks (NRN and NRW). These layers should never be modified. Depending on your project area, the layers may also contain additional surface disturbances that were mapped by the Yukon Government. You do not need to re-digitize existing features unless they are obviously wrong or they have changed in size or shape e.g., mining expansion.
- **Digitize in BP_Line and BP_Poly disturbance layers**
 - What to digitize:
 - Visible features that were not previously digitized i.e., they do not exist in the linear_disturbances and areal_disturbances layers. This includes features that are in proximity to existing (previously digitized) features or in newly disturbed areas e.g., a new mining project.
 - Existing line or polygon features that have changed in shape and size since last digitized e.g., mining expansion. Digitize the entire feature that has changed, not just the additions.
 - Existing line or polygon features that do not appear to represent the feature correctly e.g., errors or features that were previously digitized using older and coarser imagery.
- **Tracking progress using the 1-km² grid**
 - Use the 1-km² grid provided in the data package to guide your progress. One approach is to start at the top row and work from left to right, then move on to the next row, and so on. A value of 1 can then be entered in the “status” field of the grid layer to indicate which grid cells have been reviewed. This value can be added directly or noted on a piece of paper or text editor and added to the grid layer at the end of a session.
- **Scale of capture**
 - Digitizing should always be done at 1:5,000 using $\leq 1.5\text{m}$ resolution satellite imagery e.g., a SPOT mosaic or ESRI/Google imagery. Simply select the correct scale from the Scale drop-down box at the bottom of the screen.
- **Digitize features completely**
 - Start in the first 1-km² grid cell and digitize all disturbances that can be seen.
 - If the feature being digitized leaves the grid cell (such as a road), digitize it completely then return to the original grid cell and digitize the next feature.
 - The same procedure can be used if the feature you are digitizing crosses the project AOI, unless you know that someone else has already digitized the feature.

- Features should only be divided into separate entries when an attribute changes, such as the underlying imagery date, the width of a road, or if a trail turns into a road.
- **Identifying disturbances**
 - If a potential disturbance is identified, use the **Supporting layers** to help identify the feature and decide if it is a disturbance. Toggle the different supporting layers on and off to help pinpoint any disturbances. Small disturbances may be easily missed otherwise, or natural features could be mistakenly digitized as a human disturbance. For example:
 - Sometimes rivers look like roads/trails, toggle the Water courses layer to check.
 - Sometimes fires look like cutblocks or clearings, toggle the Fires layer to check.
- **Classifying disturbances**
 - Disturbances are classified using the TYPE_INDUSTRY and TYPE_DISTURBANCE attributes found in Appendix 1.
 - Use the **Supporting layers** to help classify the disturbance. For example:
 - Linear features could be powerlines, or seismic lines, or trails/roads. Toggle the various layers to check.
 - A mine site in a quartz claim area can be assumed to be TYPE_INDUSTRY: 'Mining' and TYPE_DISTURBANCE: 'Quartz mining significant'.
 - A small settlement on a lake without road access might be identified as TYPE_INDUSTRY: 'Rural' and TYPE_DISTURBANCE: 'Camp' by checking the **Tourism layer** for any nearby fishing locations.
- **Flagging for field validation**
 - If the spatial boundary or attribute assignment of a digitized feature is uncertain, it can be flagged for validation. A value of 1 should be assigned to the FLAG field and any notes relevant to the flag should be added in the NOTES field.
 - Examples could be an old trail partially visible in the imagery that needs to be ground truthed to check its location; or a feature where the digitizer is not certain about the disturbance type and wants a second opinion. FLAGS should always have an associated NOTE describing why the feature was flagged.
- **Topology rules**
 - Polygons must not overlap other polygons where both represent current disturbances on the landscape. It's possible polygons could overlap in the future if an older disturbance is no longer considered current. Features can be classified as Historic if no longer present, or Retired if mapped incorrectly (see YG manual for additional details).
 - Lines must not overlap lines (i.e. two lines should not share multiple points on top of each other), but they can intersect and cross other line features. The only case where lines may overlap is when a historic feature (e.g., a narrow trail) is replaced by a current feature (e.g., a wide road).
 - Lines can overlap polygons and digitizing of linear features that pass through a polygon is required as this could be useful for tabulating the total length or density of linear features in the dataset.

- o Features must be single-part not multi-part. However, make sure to split a feature every time an attribute changes, including the imagery date or feature width.
- o Lines and polygons should snap to polygons where they are adjacent to each other.
- o Linear features should be collected as a polyline geometry regardless of their width.

2.2 Satellite image priority

Ideally all features should be digitized using the primary reference image which, for our projects, was the 1.5m resolution SPOT 2020-21 mosaic (GeoYukon MedRes_Update), which was available until March 31, 2024 and has been replaced by an updated SPOT mosaic. In some areas, additional permanent HighRes imagery may be available but should only be used if it is more recent than the mosaic. Another source of imagery is Very High Resolution (VHR) imagery such as ESRI and Google imagery. These can have much finer resolution (e.g. 0.5m) but do not have a consistent reference year. The basemap tiles are often made up of old and newer imagery and sometimes the reference year information is not available (e.g., ESRI basemap reference year can be seen using the info tool in ArcMap but not in QGIS). Recent, high resolution Planet Imagery is available by payment or subscription.

Box 3. Other Jurisdictions

A primary basemap or reference image must be chosen for digitizing disturbances in the AOI. In the Yukon Territory, the Yukon government maintains a SPOT imagery mosaic which is i) freely available, ii) provides a relatively consistent and recent reference year, iii) has a high resolution, and iv) provides broad spatial coverage. In other jurisdictions, users will have to select imagery based on these considerations and imagery availability. Our [disturbance datasets](#) list provides a starting point. In addition, different organizations may have, or be able to obtain, subscriptions to other very high resolution imagery sources such as Planet Imagery.

VHR data can be useful in combination with the SPOT data when searching for disturbances, and for confirming disturbance types. In some cases, disturbances may be visible in the VHR that are not visible in the SPOT data, in which case the VHR data can be used for digitizing (with the caveats mentioned below).

Digitizing should be done following the three options below, with priority given in this order (i.e. digitizing using SPOT imagery is the priority if possible):

1. **Disturbance is clearly visible in SPOT imagery:**
 - a. Digitize using the most recent SPOT mosaic (Satellites/Yukon_Composite_150cm)
 - b. Assign IMAGE_NAME, attribute as "SPOT-6/7 Mosaic (2020-2021) Planet Labs", IMAGE_RESOLUTION as "1.5", IMAGE_SENSOR as "SPOT-6/7", and IMAGE_DATE as identified from the GeoYukon website (see previous paragraph).

- c. Assign VHR_ASSIST attribute as NA because no other imagery was required to identify and map the disturbance.
- 2. **Disturbance is visible in SPOT imagery, but VHR (e.g. ESRI) is needed to identify the disturbance type, and possibly inform/confirm the outline:**
 - a. Digitize using SPOT 2020-21 mosaic (GeoYukon MedRes_Update)
 - b. Assign IMAGE_NAME attribute as "SPOT-6/7 Mosaic (2020-2021) Planet Labs", IMAGE_RESOLUTION as "1.5", IMAGE_SENSOR as "SPOT-6/7", and IMAGE_DATE as identified from the GeoYukon website (see previous paragraph).
 - c. Assign VHR_ASSIST attribute as "ESRI World Imagery" because ESRI basemap was needed to assist digitizing.
- 3. **Disturbance is clearly visible in ESRI imagery and cannot be digitized using SPOT imagery:**
 - a. Digitize using ESRI imagery
 - b. Assign IMAGE_NAME as "ESRI World Imagery". Use the info tool in ArcMap to get the RESOLUTION, SENSOR and IMAGE_DATE values. As an alternative, use the ESRI World Imagery Wayback site (<https://livingatlas.arcgis.com/wayback>) to try to identify the year of imagery.
 - c. Assign VHR_ASSIST attribute as "ESRI World Imagery" (validation step will check VHR_ASSIST is always ESRI when IMAGE_NAME is ESRI).

2.2.1 Imagery Metadata

When digitizing, imagery metadata, such as the imagery date, is required to complete the attribute table of each feature correctly. For SPOT imagery accessed through GeoYukon (<https://mapservices.gov.yk.ca/GeoYukon/>), the only way to access imagery metadata is to turn on the "Medium Resolution Satellites - Current" layer on the GeoYukon website, press the "Identify Imagery" button at the top of the map, and then click on the location where the newly digitized feature is located (you will need to zoom in to the appropriate area, guided by the imagery). The panel on the left will indicate the year of imagery as well as the sensor and resolution. ArcMap's "Identify" tool must be used to access metadata about the ESRI basemap.

2.3 Digitizing with QGIS

See the QGIS [documentation \(section 5.1\)](#) for more details.

Digitizing a polygon:

1. Right click on the **Attributes Toolbar** at the top of the QGIS window and make sure the **Digitizing Toolbar** is selected.
2. Click on the layer name (i.e., BP_Poly) in the **Layers** menu to select it, then click on **Toggle Editing**, either in the **Digitizing Toolbar** or by right clicking on the layer name in the **Layers** menu. This makes the layer editable which allows us to digitize new disturbance polygons.

3. Click on **Add Polygon Feature** in the **Digitizing Toolbar**. You can now draw the feature.
4. When the feature is complete, right click on the map. A window will open where you can fill in the attributes for the new feature. Fill in the attributes then click **OK**. [Alternatively, you can leave the attributes NULL and fill them in later by right clicking on the layer name and selecting **Open Attribute Table**; attributes of all features can be edited directly in the table as long as the layer is editable].
5. Save edits regularly by clicking **Save Layer Edits** in the **Digitizing Toolbar**.
6. When digitizing is complete, turn off editing for the layer by clicking the **Toggle Editing** button again.

Digitizing a line:

1. Same as digitizing polygons but select the line layer (i.e., BP_Line) to edit.
2. The line attribute table has an additional attribute **WIDTH_M**. Estimate the width of the linear feature in metres, and rounded to the nearest whole number. Use the **Measure Line** tool in the **Attributes Toolbar** for this. Set the distance to metres and click across the width of the road. You will likely have to zoom in on the map to do this. If the width of the feature varies, take the average of several measurements.

Note that when you save the project file at the end of a work session you will also maintain the current map view.

2.4 Which features to map?

Each project area is divided into a grid consisting of 1) 100-km² cells and 2) 4-km² cells. Both grids include a binary attribute called “Priority”, which indicates the cells that should be evaluated and mapped. The larger grids can be used to track overall progress and record notes (Google spreadsheet) while smaller grids can be used to help guide the digitizing progress. Only 4-km² grid cells with a Priority value of 1 need to be reviewed and digitized. These are the areas that either have never been mapped or have been recently mapped using older SPOT data.

There are several possible scenarios:

1. The area is unmapped. In this case, any features detected need to be digitized (DATABASE = “Most Recent”).
2. The area has been mapped correctly i.e., no new features or changes to existing features can be detected using the primary reference imagery. In this case, nothing needs to be done.
3. The area has been mapped but:
 - a. Not all features have been mapped i.e., some features that can be detected using SPOT 2020-21 imagery were previously missed or did not exist on older SPOT imagery. In this case, the features need to be digitized (DATABASE = “Most Recent” in the BP_Line or BP_Poly layer).
 - b. Some features have changed in size and/or shape i.e., a feature that was correctly mapped using older imagery has since changed size and/or shape. In this case, the features need to be digitized completely (DATABASE = “Most Recent” in the BP_Line or BP_Poly layer).

BP_Poly layer) and the older feature re-attributed (Database = “Historic” in the linear_disturbances or areal_disturbances layer).

- c. Some features have been incorrectly mapped using the older SPOT imagery. In this case, the features need to be re-digitized completely (DATABASE = “Most Recent” in the BP_Line or BP_Poly layer) and the older feature re-attributed (Database = “Retired” in the linear_disturbances or areal_disturbances layer).

In all cases make sure to:

- Keep the original feature in the linear_disturbances or areal_disturbances layer i.e., do not delete the original.
- Digitize the feature needing to be remapped in the BP_Line or BP_Poly layer.
- Add the REF_ID from linear_disturbances or areal_disturbances of the feature being remapped to the FLAG_YG field in the BP_Line or BP_Poly layer.

For further information, see pages 21-24 in the YG manual.

2.5 Validating mapped features

Disturbance mapping validation can be completed in several ways depending on the resources available for the project. Ideally, all validation techniques can be applied. They are as follows:

- a) Review of digitized features by another team member
- b) Remote validation using higher resolution imagery than what was used for digitizing
- c) Field-based validation (i.e. ground truthing) by traveling features of interest in-person, using equipment such as drones, or interviewing community members.

The purpose of the validation process is to ensure features were digitized properly and were assigned the correct attributes. Validation may also reveal systematic errors that can be corrected or accounted for during analyses. More resource intensive methods such as field-based validation may prioritize features that were flagged during the digitizing process.

3. Developing a temporal disturbance database

A temporal disturbance database contains a year or date of occurrence for every feature. In our case this data is stored in the DIST_YR attribute. This is the date that the disturbance appeared on the landscape and is different to the IMAGE_DATE which is the year the satellite imagery was taken. In the absence of an accurate DIST_YR, the IMAGE_DATE is the best information we have about when a disturbance occurred, but in almost all cases the DIST_YR will be earlier than the IMAGE_DATE.

We have developed two methods for assigning dates to disturbance features according to differing project needs and available resources. The first, more resource intensive approach is detailed here and in our case study². The second, simplified approach is detailed in the temporal mapping manual on GitHub.

3.1 Identifying disturbance dates

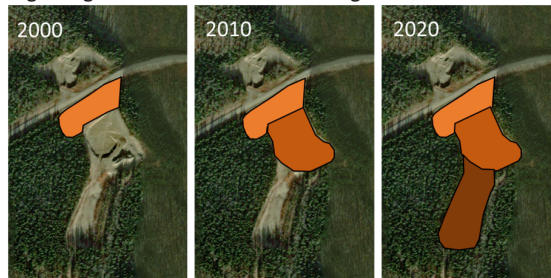
3.1.1 Manual dating method

The manual method for assigning disturbance dates is to use yearly imagery to identify when a disturbance shows up on the landscape. Since many areal disturbances will grow incrementally over time, older polygons must be digitized. The older polygons may then be subtracted from the current polygon to get a set of non-overlapping polygons with the correct years assigned, or overlapping polygons from different years may be left until the analysis stage of a project. The following procedure can achieve this:

1. Setup
 - a. Open a GIS project with the areal and linear disturbance layers
 - b. Add yearly imagery for all available years of interest. Set metadata symbology (hollow) and add labels (year acquired). Some of the imagery may require additional processing, such as merging tiles or building a virtual raster, or adjusting the appearance settings under symbology.
 - c. Arrange the yearly imagery in chronological order in the GIS project. Create folders for each decade (1980s, 1990s and so on), and load imagery in five year intervals if possible.
 - d. Ensure current line and polygon files are prepped. Imagery and sensor names of the older imagery should be added to drop down lists, the DIST_YEAR and FLAG fields should be added, and RETIRED features should be filtered out so they are not assigned dates during this stage.
 - e. Load ancillary data. Load Land Tenure layers and a Historical Mineral Claims layer, if available. Set the disposition date as a label; if unavailable, use the start date.
2. Search
 - a. Use 100 km and 1 km grids to guide you, and track your progress in the digitizing spreadsheet. Use other available resources such as Google Earth Pro (see Alternate Dating Methods, below) to help identify the time frame when a disturbance appeared. Use the additional imagery and ancillary layers in QGIS to try to narrow down the exact year the disturbance appeared.
 - b. For each disturbance feature, toggle the yearly imagery until you find the year in which the feature appears on the landscape. If the full current feature appears in that year, assign the year of that feature as the DIST_YR and move on to the next feature. Only edit the attribute table using QGIS (not Google Earth Pro).
 - c. If the feature is present in all images, then it must be older than the available imagery. A DIST_YR cannot be assigned in these cases unless older imagery or an alternate dating method is used. In the absence of older imagery, in the DIST_YR field enter a standard notation such as “-2000” to indicate the feature is older than 2000, for example.

- d. If feature dates cannot be addressed or need to be revisited, flag with a “2” in the FLAG field and add a note explaining the issue. If the feature can no longer be seen due to very poor imagery resolution but you suspect it was still present, note the last year the feature was visible with a “-” (e.g., -1990, see step 2.c. above) and then flag the feature with a “2”.
 - e. Once a 100 km grid is complete, change the status field to “2” and mark as done in the digitizing spreadsheet.
3. Digitize
- a. Option 1: The goal of this approach is to maintain a **flat dataset** as shown in Figure 1. If only a portion of a feature was present in a given year, the disturbed area for that year should be digitized and then removed from the present day polygon using the ArcGIS Erase tool or QGIS Difference tool. Some features (e.g. mine sites) may end up being composed of many polygons each with different DIST_YR values. Previous versions of a feature (such as a gravel pit that has grown over time) should be set to “Historic” under the DATABASE field.

Digitizing forward in time as new images become available



Digitizing backward in time using archived imagery

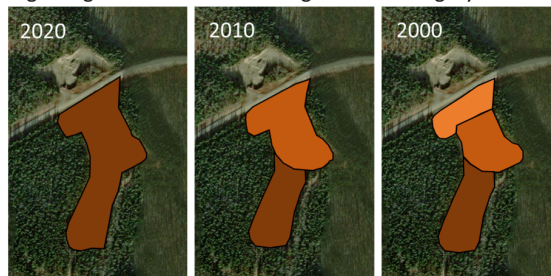


Figure 1. Examples of digitizing forward in time as new imagery becomes available, vs digitizing backward in time using archived images. In the above example we assume that the disturbance feature grew over the three timesteps (i.e. in the top left image, the orange polygon should represent the full extent of the disturbance in 2000). When digitizing forward, only the newly added sections of disturbance are digitized. When moving backward, the 2020 polygon needs to be split into three polygons, each with the correct date assigned.

- b. Option 2: This approach will create a series of **overlapping lines and/or polygons** in some areas. Digitize the feature in its entirety each time it changes, recording the disturbance year and other attributes accordingly, including “Historic” in the DATABASE

field. Prior to analysis the features can be filtered to the desired disturbance year and merged to create a separate flat dataset.

3.1.2 Alternate dating methods

For disturbances that cannot be dated using the method above, some alternatives are listed below. These methods do not permit the digitizing step described above so are only valid if the entire feature can be assigned a single DIST_YR value.

1. Imagery that cannot be loaded into the GIS project. e.g. Google Timelapse.
 - a. Google Timelapse and Google Earth Pro have a timeline feature that allows users to scroll through years to see changes. These tools can be helpful to find the year a feature appears, then switch to the GIS project to update the feature's attribute table. One advantage of this approach is that these tools have cloud free images which are not always available by accessing other imagery sources. A .kml file of the current line and polygon layers can be uploaded to Google Earth Pro to assist with identifying features.
 - b. Users can also digitize older disturbance features using Google Earth Pro's timelapse imagery. This is not an optimal workflow because switching between two GIS platforms is slow and could introduce unforeseen bugs. That said, the workflow would be:
 - i. Import disturbance .shp into Google Earth Pro. It will be converted to a .kml.
 - ii. Digitize as needed.
 - iii. Convert back to shapefile using ESRI toolbox or QGIS equivalent.
 - iv. Reassign attributes in QGIS/ARCGIS, as Google Earth Pro does not preserve the attribute table.
2. Government records; these may often be dates of licenses and not necessarily disturbance dates.
 - a. <https://yesabregistry.ca/>
 - b. <https://mapservices.gov.yk.ca/GeoYukon/>
 - c. <https://catalogue.data.gov.bc.ca/>
3. Online research e.g. this [fishing camp](#) website describes the history and building of the lodge.
4. Validation by local knowledge holders.

3.2 Limitations

Assigning disturbance dates is easier for more recent disturbances than for older ones because older imagery has reduced quality and availability. In older imagery, it's much easier to assign dates to larger features such as mines, cutblocks etc. Some roads and smaller features will be impossible to date according to imagery because they don't show up in older imagery (i.e. LANDSAT @ 30m). In these cases, alternative methods such as validation by local knowledge holders, reviewing historic documents, and extrapolating from dates of nearby disturbances may be used.

Appendix 1. Industry and disturbance types

Industry and disturbance types are taken from the Anthropogenic Surface Disturbance Mapping in the Yukon, Standards and Guidelines for Contractors, v 4.0.2 (Tables 4 and 7). Examples of each disturbance type can be seen in Anthropogenic Surface Disturbance Mapping in the Yukon, Feature Interpretation Key for Contractors, v 1.0.2. **Note 1.** Placer mining - major vs minor: minor placer mines are single, smaller mining projects in isolation, while major placer mines cover an entire valley or region and are typically composed of multiple adjacent mines. **Note 2.** Trails vs. unpaved roads: unpaved roads have been developed and are wide enough for a truck to go down. Trails are undeveloped and narrow enough that a truck wouldn't fit down them. Note (June 2026): The Yukon Government is updating their user manuals and standards which could potentially result in some changes to the attributes listed below.

Industry type	Feature type	Disturbance type	Description
Agriculture	Areal	Agriculture	Farms, ranches, or other agricultural areas
Forestry	Areal	Forestry	Cut blocks or other forestry related activities
Mining	Areal	Building	A building footprint or the building and the surrounding land related to mining activities
		Drill Pad	Drill pad features related to mineral exploration activities
		Fuel Cache	Remote caches of fuel allowing for mineral exploration activities (will often have fuel tanks and barrels)
		Gravel Pit / Quarry	Pit or quarry for mining gravel or aggregate
		Laydown area	Areas used to store materials and equipment for mining operations
		Mining	Miscellaneous or unknown mining activities
		Placer Mining - Minor	Placer mining area with little disturbance
		Place Mining - Significant	Placer mining area with greater disturbance
		Quartz Mining - Minor	Quartz mining area with little disturbance
		Quartz Mining - Significant	Quartz mining area with greater disturbance
		Tailing Pond	Tailing pond associated with mining activity
		Camp	Mining camp
	Linear	Diversion Channel	An engineered linear bypass feature that redirects a watercourse around a mining feature such as a tailings pond
		Survey / Cutline	A linear cleared area through undeveloped land, used for line-of-sight surveying; impossible to distinguish whether associated with quartz or placer mining (overlapping or unclear claims information)
		Survey / Cutline - Placer	A linear cleared area through undeveloped land, used for line-of-sight surveying; associated with placer mining (identified using claims information and/or other indicators)
		Survey / Cutline - Quartz	A linear cleared area through undeveloped land, used for line-of-sight surveying; associated with quartz mining (identified using claims information and/or other indicators)
		Trench	A long, narrow excavation dug to expose vein or ore structure
		Unknown	Unknown linear mining disturbance
Oil and Gas	Areal	Well Pad	Cleared area surrounding oil or gas well
		Unknown	Added by BEACONS 2024-02-27
	Linear	Pipeline	Visible pipeline or pipeline Right-of-Way (above- or below-ground)

Industry type	Feature type	Disturbance type	Description
Agriculture	Areal	Agriculture	Farms, ranches, or other agricultural areas
Forestry	Areal	Forestry	Cut blocks or other forestry related activities
Mining	Areal	Building	A building footprint or the building and the surrounding land related to mining activities
		Drill Pad	Drill pad features related to mineral exploration activities
		Fuel Cache	Remote caches of fuel allowing for mineral exploration activities (will often have fuel tanks and barrels)
		Gravel Pit / Quarry	Pit or quarry for mining gravel or aggregate
		Laydown area	Areas used to store materials and equipment for mining operations
		Mining	Miscellaneous or unknown mining activities
		Placer Mining - Minor	Placer mining area with little disturbance
		Place Mining - Significant	Placer mining area with greater disturbance
		Quartz Mining - Minor	Quartz mining area with little disturbance
		Quartz Mining - Significant	Quartz mining area with greater disturbance
		Tailing Pond	Tailing pond associated with mining activity
		Camp	Mining camp
	Linear	Diversion Channel	An engineered linear bypass feature that redirects a watercourse around a mining feature such as a tailings pond
		Survey / Cutline	A linear cleared area through undeveloped land, used for line-of-sight surveying; impossible to distinguish whether associated with quartz or placer mining (overlapping or unclear claims information)
		Seismic Line	Seismic lines
Rural	Areal	Camp	Any camp outside of mining areas, including fishing/hunting camps, ENV conservation officer cabins/camps, outfitters, etc.
		Homestead	Rural dwelling and associated land
	Linear	Driveway	A driveway in a rural area
		Fence	A fence in a rural area
Transportation	Areal	Airstrip	Airport or Airstrip
		Clearing	Clearings that are related to transportation but could not be clearly attributed as a turn area, pullout, road cut and fill, etc.
		Gravel Pit / Quarry	Gravel pits related to transportation
		Pullout / Turn Area	An area associated with transportation and is intended as a vehicle pullout or turn area
		Road Cut and Fill	Cut slopes and moved earth for road construction purposes
		Unknown	Added by BEACONS 2023-11-28
	Linear	Access Assumed	A linear feature that is assumed to be an access road, but could also be a trail
		Access Road	A road or narrow passage whose primary function is to provide access for resource extraction (i.e. mining, forestry) and may also have served in providing public access to the backcountry.
		Arterial Road	A major thoroughfare with medium to large traffic capacity
		Local Road	A low-speed thoroughfare, provides access to front of properties, including those with potential public restrictions such as trailer parks, First Nations land, private estate, seasonal residences, gravel pits (NRN definition for Local Street/Local Strata/Local Unknown). Shows signs of regular use.
		Right of Way	For Road Rights as attributed in the land parcels ancillary data

Industry type	Feature type	Disturbance type	Description
Agriculture	Areal	Agriculture	Farms, ranches, or other agricultural areas
Forestry	Areal	Forestry	Cut blocks or other forestry related activities
Mining	Areal	Building	A building footprint or the building and the surrounding land related to mining activities
		Drill Pad	Drill pad features related to mineral exploration activities
		Fuel Cache	Remote caches of fuel allowing for mineral exploration activities (will often have fuel tanks and barrels)
		Gravel Pit / Quarry	Pit or quarry for mining gravel or aggregate
		Laydown area	Areas used to store materials and equipment for mining operations
		Mining	Miscellaneous or unknown mining activities
		Placer Mining - Minor	Placer mining area with little disturbance
		Place Mining - Significant	Placer mining area with greater disturbance
		Quartz Mining - Minor	Quartz mining area with little disturbance
		Quartz Mining - Significant	Quartz mining area with greater disturbance
		Tailing Pond	Tailing pond associated with mining activity
		Camp	Mining camp
	Linear	Diversion Channel	An engineered linear bypass feature that redirects a watercourse around a mining feature such as a tailings pond
		Survey / Cutline	A linear cleared area through undeveloped land, used for line-of-sight surveying; impossible to distinguish whether associated with quartz or placer mining (overlapping or unclear claims information)
		Trail	Path or track used for walking, cycling, ORV, or other backcountry activities. (Note: trails used for mining activities are Access Roads.)
		Unpaved Road	Dirt or gravel road that does not necessarily access remote resources
		Railway	Added by BEACONS 2023-11-28
		Unknown	Added by BEACONS 2023-11-28
Unknown	Areal	Clearing	A tract of land devoid (or nearly devoid) of natural land cover and suspected to be anthropogenic in nature
		Gravel Pit / Quarry	A gravel pit with unknown related industry
		Unknown	Unable to identify from imagery, but suspected to be anthropogenic
	Linear	Right of Way	A right of way with unknown industry type
		Survey / Cutline	A linear cleared area through undeveloped land, used for line-of-sight surveying. A cutline may not always be associated with mineral exploration, therefore, Type: Unknown was used to differentiate all cutlines that were outside of mineral exploration.
		Unknown	Unclassified, or unable to identify type based on imagery, but suspected to be anthropogenic
Urban	Areal	Building	Visible building or structure
		Cemetery	Cemetery
		Clearing	Miscellaneous urban clearings
		Cul-de-sac / Turn Area	A turn area associated with transportation or road cul-de-sac
		Dam	Barrier impounding water or stream
		Golf Course	Recreational golfing area
		Industrial	Areas that are designated for industrial uses: factories, tank farm, transportation area
		Institutional	Any institutional buildings and immediate cleared area: School, government, etc.

Industry type	Feature type	Disturbance type	Description
Agriculture	Areal	Agriculture	Farms, ranches, or other agricultural areas
Forestry	Areal	Forestry	Cut blocks or other forestry related activities
Mining	Areal	Building	A building footprint or the building and the surrounding land related to mining activities
		Drill Pad	Drill pad features related to mineral exploration activities
		Fuel Cache	Remote caches of fuel allowing for mineral exploration activities (will often have fuel tanks and barrels)
		Gravel Pit / Quarry	Pit or quarry for mining gravel or aggregate
		Laydown area	Areas used to store materials and equipment for mining operations
		Mining	Miscellaneous or unknown mining activities
		Placer Mining - Minor	Placer mining area with little disturbance
		Place Mining - Significant	Placer mining area with greater disturbance
		Quartz Mining - Minor	Quartz mining area with little disturbance
		Quartz Mining - Significant	Quartz mining area with greater disturbance
		Tailing Pond	Tailing pond associated with mining activity
		Camp	Mining camp
	Linear	Diversion Channel	An engineered linear bypass feature that redirects a watercourse around a mining feature such as a tailings pond
		Survey / Cutline	A linear cleared area through undeveloped land, used for line-of-sight surveying; impossible to distinguish whether associated with quartz or placer mining (overlapping or unclear claims information)
		Landfill	Site used for disposal of waste materials
		Pond	Standing body of water, created anthropogenically; includes sewage lagoons, wastewater facilities, and artificial bodies of water
		Recreation Area	Visible disturbance in Urban / Rural parks and recreation areas
		Rural Residential	Land use in which housing predominates in an urban or community setting
		Tower	A tall structure, possibly used for communications or forestry
		Urban	Miscellaneous or unknown urban features
Utility	Linear	Electric Utility Corridor	Corridor usually running parallel to highway, where transmission lines or other utilities are visible
		Unknown	Unknown linear feature assumed to be a utility corridor; ancillary data is unclear